

Comparison of data warehouses and data lakehouses

So far, you've learned about the common ways that data is stored. Specifically, you've explored data warehouses and data lakes. As a refresher, data warehouses are databases that consolidate data from multiple source systems for consistency, accuracy, and efficient access. Data lakes are storage systems that store large amounts of raw data in its original format until it's needed. This could include structured, semi-structured, and unstructured data. In this reading, you'll learn about another data storage solution: data lakehouses. You'll also learn more about how data lakehouses compare to data warehouses.

Data warehouses vs data lakehouses

Data warehouses are usually highly efficient, but they also have limited scalability and can be expensive to maintain. Data lakes have more flexible storage, but are usually large and complex. To benefit from each storage system, and overcome their drawbacks, data lakehouses combine both systems. Because of this, data lakehouses have a lot in common with data warehouses. But, there are some key differences, too. Review the chart for an overview of the benefits and challenges of data warehouses and data lakes:

Storage solution	Benefits	Challenges
Data Warehouse	• Able to gather data from multiple sources into a centralized location • Easy to query and report data from within a database • Highly secure • Has consistent data quality and integrity that is compliant with organizational standards	• Not as flexible as other data storage solutions because it requires structured data • Expensive to implement and maintain • Difficult to retain data since historical data is resource-heavy to maintain • Often separate from operational systems, creating additional complex processing needs

Data Lakehouse	• Greater flexibility with the storage of structured and unstructured data • Centralized data storage across data types • Cost effective implementation and maintenance • Can adopt ML and AI solutions	• Data can require more processing to be useful • Higher learning curve for users interacting with the system • More difficult to enforce data compliance
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Key takeaways

Data storage is a key part of working with data, and is just as important for cloud data analytics professionals as it is for traditional data analysts. As you continue to learn more about the different data storage solutions, and how they can be used together, consider how each system can contribute to a larger cloud architecture. For example, data warehouses and lakehouses have benefits and challenges, but they can also be used together as a way to take advantage of their benefits, and make up for their shortcomings.